

Pizza Sales Analysis Using SQL

1. Executive Summary

This project analyzes pizza sales data using SQL. The main goal is to understand customer ordering behavior, sales performance, and revenue trends. By analyzing the data, valuable business insights were generated to support better decision-making.

2. Business Problem

The pizza company collects a large amount of sales data every day. However, it is difficult to understand business performance without proper analysis. This project helps identify top-selling pizzas, revenue-generating products, customer preferences, and sales patterns using SQL.

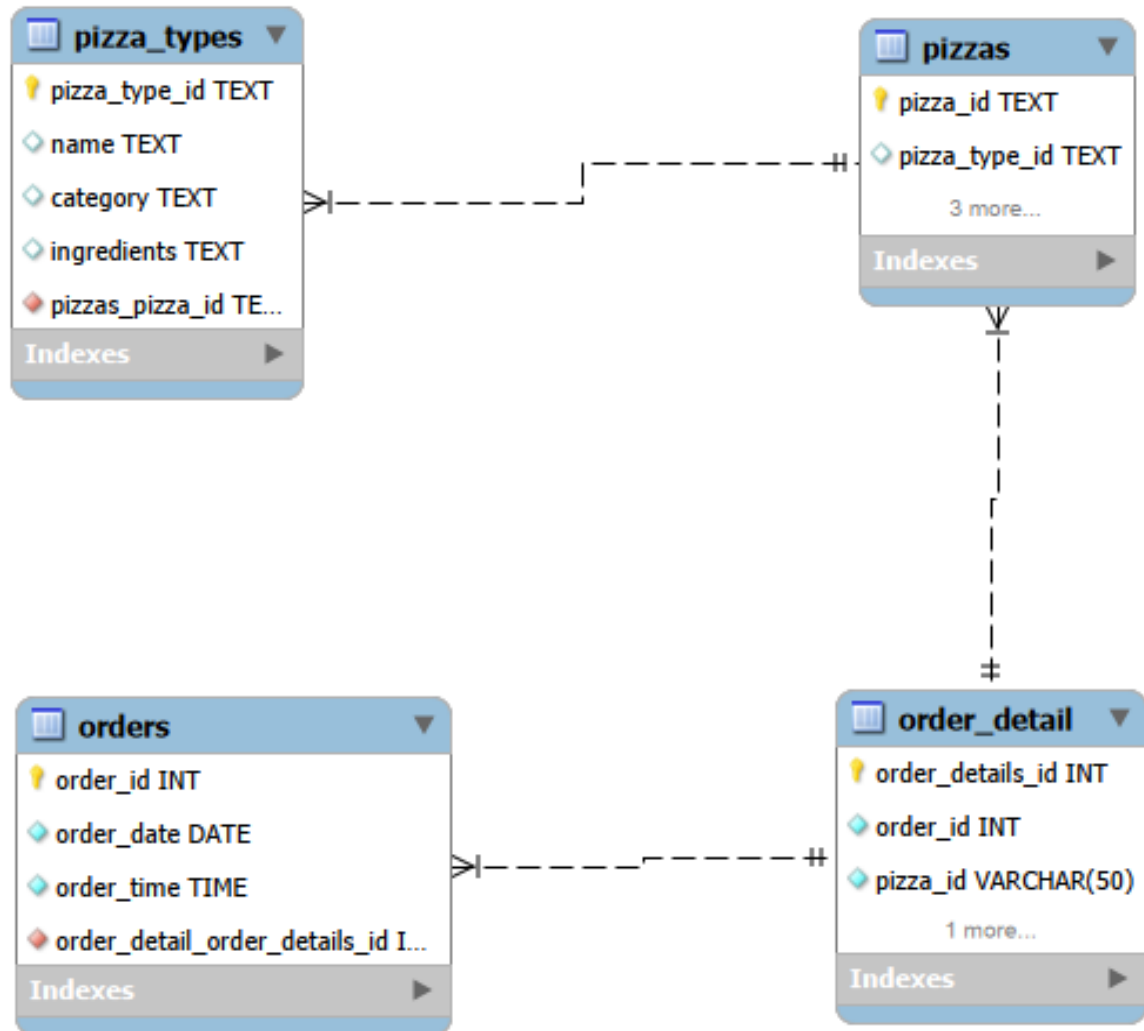
3. Dataset Overview

The dataset contains four tables:

- **Orders**
Stores information about order date and order time.
 - **Order Details**
Stores information about pizzas ordered and their quantities.
 - **Pizzas**
Contains pizza size and price information.
 - **Pizza Types**
Contains pizza names, categories, and ingredients.
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4. Database Schema

The following ER Diagram illustrates the relationships between the Orders, Order Details, Pizzas, and Pizza Types tables used in the analysis.



Legend

□ Primary Key (PK)

□ FK Foreign Key

1 One

< Many

Relationships

pizza_types (1) --- pizzas (M)

pizzas (1) --- order_detail (M)

orders (1) --- order_detail (M)

5. Data Model Explanation

The database consists of four related tables.

- One pizza type can have multiple pizza sizes.
- One pizza can appear in many orders.
- One order can contain multiple pizzas.

Relationships

- Pizza Types (1) → Pizzas (Many)
 - Pizzas (1) → Order Details (Many)
 - Orders (1) → Order Details (Many)
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6. SQL Concepts Used

The following SQL concepts were used in this project:

- SELECT Statement
- WHERE Clause
- GROUP BY
- ORDER BY
- Aggregate Functions
- INNER JOIN
- Common Table Expressions (CTE)
- Window Functions

These concepts helped perform data analysis and generate business insights.

7. Business Questions & Analysis

Q1 Retrieve the total number of orders placed.



```
1 select count(*) from orders;
```

Business Insight:

The company received **21350** orders during the analysis period, indicating the overall level of customer demand and business activity.

Q2 Calculate the total revenue generated from pizza sales.



```
1 SELECT
2     round(SUM(order_detail.quantity * pizzas.price),2) AS 'Total Revenue'
3 FROM
4     order_detail
5     JOIN
6     pizzas ON order_detail.pizza_id = pizzas.pizza_id;
```

Business Insight:

The company generated **\$ 817860.05** in total revenue, demonstrating the overall financial performance of the business.

Q3 Identify the highest-priced pizza.

```
1 SELECT
2     pizza_types.name, pizzas.price AS 'Highest Price'
3 FROM
4     pizza_types
5     JOIN
6     pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
7 ORDER BY price DESC
8 LIMIT 1;
```

Business Insight:

The Greek Pizza was identified as the highest-priced pizza on the menu, representing a premium product offering.

Q4 Identify the most common pizza size ordered.

```
1 SELECT
2     pizzas.size, COUNT(order_detail.order_details_id)
3 FROM
4     pizzas
5     JOIN
6     order_detail ON pizzas.pizza_id = order_detail.pizza_id
7 GROUP BY pizzas.size;
```

Business Insight:

Customers most frequently ordered the pizza size are:-

	size	Total_Orders
▶	M	15385
	L	18526
	S	14137
	XL	544
	XXL	28

Q5 List the top 5 most ordered pizza types along with their quantities.

```
1  SELECT
2      pizza_types.name, SUM(order_detail.quantity) as quantity
3  FROM
4      pizza_types
5      JOIN
6      pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
7      JOIN
8      order_detail ON pizzas.pizza_id = order_detail.pizza_id
9      group by pizza_types.name
10 ORDER BY quantity DESC
11 LIMIT 5;
```

Business Insight:

The top-selling pizza types are:-

	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Q6 Determine the distribution of orders by hour of the day.

```
1  SELECT
2      HOUR(order_time) AS Hours,
3      COUNT(order_id) AS Quantity
4  FROM
5      orders
6  GROUP BY Hours
7  ORDER BY Quantity desc;
```

Business Insight:

Order activity peaked during

	Hours	Quantity
▶	12	2520
	13	2455
	18	2399
	17	2336
	19	2009
	16	1920

Q7 Join the necessary tables to find the total quantity of each pizza category ordered.

```
1  SELECT
2      pizza_types.category, SUM(order_detail.quantity) AS Quantity
3  FROM
4      pizza_types
5      JOIN
6      pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
7      JOIN
8      order_detail ON pizzas.pizza_id = order_detail.pizza_id
9  GROUP BY pizza_types.category
10 ORDER BY Quantity DESC;
```

Business Insight:

The **Classic** category recorded the highest sales volume, making it the most popular category among customers.

Q8 Join relevant tables to find the category-wise distribution of pizzas.

```
1  SELECT
2      category, COUNT(name)
3  FROM
4      pizza_types
5  GROUP BY category;
```

Business Insight:

The menu is distributed across multiple categories

	category	COUNT(name)
►	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

Q9 Group the orders by date and calculate the average number of pizzas ordered per day.

```
1 SELECT
2     ROUND(AVG(quantity), 0) AS avg_pizza_order
3 FROM
4     (SELECT
5         orders.order_date, SUM(order_detail.quantity) AS quantity
6     FROM
7         orders
8     JOIN order_detail ON orders.order_id = order_detail.order_details_id
9     GROUP BY orders.order_date) AS order_quantity;
```

Business Insight:

On average, customers ordered **61** pizzas per day, reflecting the daily sales performance of the business.

Q10 Determine the top 3 most ordered pizza types based on revenue.

```
1  SELECT
2      pizza_types.name,
3      round(SUM(order_detail.quantity * pizzas.price)) AS Total_Revenue
4  FROM
5      pizza_types
6      JOIN
7      pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
8      JOIN
9      order_detail ON order_detail.pizza_id = pizzas.pizza_id
10     group by pizza_types.name
11     order by Total_Revenue desc
12     limit 3;
```

Business Insight:

The top revenue-generating pizzas are:-

	name	Total_Revenue
▶	The Thai Chicken Pizza	43434
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41410

8. Key Insights

The analysis revealed several important business insights:

- Some pizza categories generated higher sales than others.
 - Customer demand varied across pizza sizes.
 - A few pizza types contributed significantly to total revenue.
 - Sales patterns showed clear customer preferences.
 - Revenue was driven by a small group of top-performing products.
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9. Business Recommendations

Based on the analysis, the following recommendations can help improve business performance:

- **Improve Inventory Management**
Maintain sufficient stock for high-demand pizzas.
 - **Promote Best-Selling Products**
Focus marketing efforts on top-performing pizzas.
 - **Optimize Menu Offerings**
Review low-performing products and improve menu strategy.
 - **Plan Resources Efficiently**
Use sales trends to improve staffing and operations planning.
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10. Conclusion

This project successfully analyzed pizza sales data using SQL to identify customer preferences, sales trends, and revenue drivers. The insights generated can help improve inventory planning, menu optimization, and business decision-making. The project also demonstrates practical SQL skills used in real-world data analysis.

11. Skills Demonstrated

Technical Skills

- SQL
- Data Analysis
- Database Design
- Data Aggregation
- Query Writing
- Business Intelligence

Analytical Skills

- Problem Solving
 - Revenue Analysis
 - Trend Analysis
 - Business Insight Generation
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